

ALEXANDRIA ESLE, LA, USA

WMO: 722487

Lat: 31.395N

Lon: 92.291W

Elev: 118

StdP: 14.63

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 13935

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 25.9	(c) 28.5	(d) 14.5	(e) 11.5	(f) 35.9	(g) 18.9	(h) 14.3	(i) 38.8	(j) 18.5	(k) 66.0	(l) 16.6	(m) 64.6	(n) 1.6	(o) 30	(p) 0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	97.9	76.7	95.7	77.1	93.6	77.0	80.4	90.1	79.7	89.8	79.0	88.9	6.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.2	147.3	83.6	77.3	142.5	83.2	76.4	138.5	82.7	44.0	90.2	43.1	89.8	42.4	88.9	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.2	(b) 13.1	(c) 11.7	(e) 20.1	(f) 100.7	(g) 3.6	(h) 3.6	(i) 17.5	(j) 103.3	(k) 15.4	(l) 105.4	(m) 13.4	(n) 107.4	(o) 10.8	(p) 110.0		
(a) 16.2	(b) 13.1	(c) 11.7	(e) 18.8	(f) 82.2	(g) 3.5	(h) 1.6	(i) 16.3	(j) 83.4	(k) 14.2	(l) 84.3	(m) 12.3	(n) 85.2	(o) 9.8	(p) 86.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.8	49.3	53.1	59.9	66.5	74.2	80.1	82.6	82.5	77.5	67.1	57.0	50.8
	DBStd	14.18	10.62	9.80	8.85	7.27	5.80	3.75	3.05	3.49	5.23	7.99	8.89	9.95
	HDD50	382	145	70	19	1	0	0	0	0	0	1	34	112
	HDD65	1931	497	351	207	69	8	0	0	0	2	74	268	455
	CDD50	6508	124	158	325	495	750	902	1009	1008	826	531	243	137
	CDD65	2583	11	19	48	113	294	452	544	543	378	139	27	15
	CDH74	25964	50	98	348	1057	2862	4630	5860	5811	3590	1381	210	67
	CDH80	11977	7	8	46	245	1132	2252	3005	3087	1704	460	27	4
(14) Wind	WSAvg	4.2	4.8	5.3	5.3	5.5	4.5	3.7	3.3	3.2	3.0	3.3	3.8	4.5
(15) Precipitation	PrecAvg	53.0	5.2	4.2	5.2	4.3	5.1	2.7	5.0	3.9	3.8	3.0	3.9	6.5
	PrecMax	76.0	11.9	7.9	12.0	8.9	11.1	7.0	10.0	7.6	9.5	9.7	8.8	13.1
	PrecMin	33.1	1.2	1.9	1.0	0.3	1.7	0.5	1.7	0.4	1.2	0.3	0.1	2.2
	PrecStd	10.8	2.8	2.0	3.2	3.0	2.6	2.0	2.2	2.0	2.4	2.7	2.3	3.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.4	81.2	84.0	87.7	93.4	99.2	100.1	101.1	97.8	91.6	83.6	79.7
		MCWB	66.3	68.1	68.0	72.0	74.7	76.6	76.9	76.5	75.1	72.9	70.6	69.4
	2%	DB	74.3	77.2	81.2	85.2	90.6	95.8	97.1	98.1	94.5	88.2	79.4	75.5
		MCWB	64.6	65.3	67.1	69.8	74.1	76.9	77.3	76.8	75.7	72.7	68.1	67.8
	5%	DB	71.0	73.4	78.3	83.0	88.7	93.1	94.9	95.6	91.9	85.2	76.0	72.1
		MCWB	63.8	65.0	65.6	69.2	74.0	76.8	77.6	77.5	75.7	71.0	65.9	66.1
	10%	DB	67.2	70.2	74.8	80.5	86.6	90.8	92.7	93.3	89.5	82.1	72.8	68.7
		MCWB	61.0	63.6	64.4	67.8	73.3	76.4	77.7	77.5	74.9	69.7	65.4	63.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.1	71.0	71.8	75.7	78.9	80.9	81.6	81.6	80.1	78.1	73.4	72.0
		MCDB	75.4	75.9	77.8	82.5	88.1	89.5	90.7	90.9	88.9	86.1	78.2	76.8
	2%	WB	67.8	69.1	70.0	73.5	77.2	79.6	80.3	80.4	78.7	76.2	71.2	69.6
		MCDB	72.4	74.4	76.4	80.5	85.9	89.1	90.2	90.5	87.9	83.5	75.6	74.0
	5%	WB	65.4	67.1	68.5	72.2	76.0	78.6	79.6	79.6	77.8	74.7	69.0	67.5
		MCDB	69.5	71.8	74.9	79.0	84.7	88.1	89.8	90.2	86.7	81.2	74.3	71.9
	10%	WB	62.5	64.5	66.9	70.7	74.9	77.9	78.8	78.8	76.7	72.8	66.4	64.2
		MCDB	66.2	69.1	73.1	77.3	83.4	87.4	88.7	89.0	85.2	78.4	71.6	67.8
(35) Mean Daily Temperature Range	5% DB	MDBR	21.6	21.6	22.8	23.3	22.2	20.6	20.3	21.5	22.6	25.8	24.3	21.7
		MCDBR	23.6	23.3	24.0	23.6	23.0	23.2	22.9	24.5	25.0	26.7	26.4	23.6
	5% WB	MCWBR	15.7	14.0	12.2	11.3	9.4	8.0	7.2	7.5	9.2	12.5	15.8	16.4
		MCDWR	18.8	18.7	19.6	19.3	20.0	19.9	20.2	21.0	20.6	20.4	21.2	20.4
(40) Clear-Sky Solar Irradiance	taub	taub	0.329	0.351	0.371	0.413	0.440	0.478	0.502	0.495	0.443	0.372	0.353	0.341
		taud	2.527	2.465	2.406	2.291	2.261	2.197	2.157	2.176	2.294	2.464	2.469	2.492
	Ebn at Noon	Ebn at Noon	285	288	289	279	271	259	252	253	263	277	273	274
		Edh at Noon	27	32	36	42	44	46	48	47	40	31	29	27
(44) All-Sky Solar Radiation	RadAvg	RadAvg	875	1007	1393	1727	1916	1935	1918	1807	1593	1354	1010	775
	RadStd	RadStd	78	172	114	135	126	161	141	163	144	186	102	87

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+319	N/A
(47) Regional (25 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+212	+139

Nomenclature: See separate page